

Gen AI & Agentic AI Full Roadmap

Goal

This roadmap is designed to take a student from:

Python Beginner → SQL & Data Handling → APIs → Gen AI → RAG → AI Agents → Multi-Agent Systems → Deployment

Perfect for:

- Python developers
 - MERN developers moving into AI
 - Students preparing for AI engineer roles
 - Freelancers
 - AI SaaS builders
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PHASE 1 — Python Fundamentals

Module 1: Python Basics

Topics

- Python Introduction
- Variables
- Data Types
- Input/Output
- Type Casting
- Operators
- Conditional Statements
- Loops
- Functions
- Scope
- Modules
- Exception Handling
- File Handling

Practice Projects

- Calculator
- Student Marks System
- Expense Tracker

- ATM Machine
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Module 2: Advanced Python

Topics

- OOPs
- Classes & Objects
- Inheritance
- Polymorphism
- Encapsulation
- Abstraction
- Decorators
- Iterators
- Generators
- Lambda Functions
- Map / Filter / Reduce
- List Comprehension
- Virtual Environment
- Pip

Modules to Learn

- os
- sys
- json
- datetime
- random
- math
- requests
- threading
- asyncio

Projects

- File Organizer
 - PDF Merger
 - Email Sender
 - Weather App
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PHASE 2 — SQL & Databases

Module 3: SQL Basics

Topics

- Databases Introduction
- MySQL Installation
- CREATE DATABASE
- CREATE TABLE
- INSERT
- SELECT
- UPDATE
- DELETE
- WHERE
- ORDER BY
- LIMIT
- DISTINCT

Practice Tables

- Employees
 - Customers
 - Amazon Sales
 - Students
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Module 4: Intermediate SQL

Topics

- Aggregate Functions
- GROUP BY
- HAVING
- JOINS
- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- SELF JOIN
- Subqueries
- Views
- Constraints
- CASE WHEN
- CTEs

Projects

- Employee Management
 - E-commerce Analytics
 - Sales Dashboard Queries
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Module 5: Advanced SQL

Topics

- Window Functions
- ROW_NUMBER
- RANK
- DENSE_RANK
- LEAD
- LAG
- FIRST_VALUE
- LAST_VALUE
- NTILE
- Recursive CTEs
- Stored Procedures
- Transactions
- Indexing
- Optimization
- Date Functions
- Time Functions

Real-world Tasks

- Amazon Analytics
 - Customer Retention Analysis
 - Revenue Trends
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PHASE 3 — NumPy, Pandas & Plotly

Module 6: NumPy

Topics

- Arrays
- Array Indexing
- Array Slicing
- Reshaping
- Broadcasting

- Mathematical Operations
- Random Module
- Statistics

Practice

- Student Marks Analysis
 - Sales Calculations
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Module 7: Pandas

Topics

- DataFrames
- Series
- Reading CSV/Excel
- Filtering
- Sorting
- GroupBy
- Merge
- Apply
- Missing Values
- Date Handling
- Aggregations

Real Datasets

- Amazon Sales
- Netflix Dataset
- IPL Dataset
- Employee Dataset

Projects

- Sales Analytics Dashboard
 - HR Analytics
 - Expense Analysis
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Module 8: Plotly

Topics

- Line Charts
- Bar Charts
- Pie Charts

- Scatter Plots
- Histograms
- Interactive Graphs
- Dashboards

Projects

- Sales Dashboard
 - Expense Dashboard
 - AI Analytics Dashboard
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PHASE 4 — APIs & Backend Development

Module 9: APIs Basics

Topics

- What is API
- Types of APIs
- REST APIs
- RESTful APIs
- HTTP Methods
- Status Codes
- Request & Response Lifecycle
- Headers
- Query Params
- Path Params
- JSON

Tools

- Postman
 - Swagger Docs
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Module 10: FastAPI

Topics

- FastAPI Introduction
- Routes
- GET
- POST
- PUT
- DELETE

- Request Handling
- Response Handling
- Swagger Docs
- CRUD Operations
- Validation
- Middleware
- File Upload
- Authentication
- JWT
- Async Programming

Projects

- Employee CRUD
 - Food App API
 - Expense Tracker API
 - AI Chat API
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PHASE 5 — Streamlit

Module 11: Streamlit Basics

Topics

- Streamlit Introduction
- Inputs
- Buttons
- Forms
- Tables
- Session State
- File Upload
- Sidebar
- Metrics
- Charts

Projects

- Expense Tracker
 - AI Chatbot
 - Analytics Dashboard
 - Resume Analyzer
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Module 12: Streamlit + Database

Topics

- Streamlit + MySQL
- Streamlit + Supabase
- Streamlit + Aiven
- CRUD Operations
- Authentication
- Deployments

Projects

- Expense Tracker
 - Employee Portal
 - AI Dashboard
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PHASE 6 — Git & GitHub

Module 13: Git & GitHub

Topics

- Git Basics
- Repositories
- Commits
- Branches
- Push/Pull
- Merge
- GitHub Workflow
- README.md
- Collaboration

Practice

- Push AI Projects
 - Deploy GitHub Projects
-

PHASE 7 — AI Fundamentals

Module 14: AI Basics

Topics

- What is AI
 - Types of AI
 - Narrow AI
 - General AI
 - Super AI
 - Machine Learning Basics
 - Deep Learning Basics
 - Neural Networks
 - NLP
 - Computer Vision
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Module 15: Generative AI

Topics

- What is Gen AI
- How Gen AI Works
- Transformers
- Tokens
- Embeddings
- Vector Databases
- Prompt Engineering
- Hallucinations
- Temperature
- Context Window

Practice

- Prompt Engineering Tasks
 - AI Writing Assistant
 - AI Q&A Bot
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PHASE 8 — LLMs & APIs

Module 16: LLM Fundamentals

Topics

- What are LLMs
- How LLMs Work
- Training vs Inference
- Fine-tuning Basics
- Embeddings
- Open Source vs Closed Source Models

Models to Learn

- GPT
 - Gemini
 - Claude
 - Llama
 - Mistral
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Module 17: LLM APIs

Topics

- API Keys
- Chat Completions
- Streaming Responses
- System Prompts
- Temperature
- Function Calling
- Structured Output

Providers

- OpenAI
- Google Gemini
- Anthropic
- Groq
- OpenRouter

Projects

- AI Chatbot
- AI Email Generator
- AI Resume Analyzer

PHASE 9 — RAG & Vector Databases

Module 18: RAG Fundamentals

Topics

- What is RAG
- Why RAG
- Embeddings
- Chunking
- Retrieval
- Similarity Search
- Context Injection

RAG Pipeline

```
graph TD; A[User Query] --> B[Embedding Model]; B --> C[Vector Database Search]; C --> D[Relevant Chunks]; D --> E[LLM]; E --> F[Final Response];
```

Module 19: Vector Databases

Topics

- Pinecone
- ChromaDB
- FAISS
- Weaviate
- Supabase Vector

Projects

- PDF Chatbot
 - YouTube Q&A Bot
 - Company Knowledge Assistant
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PHASE 10 — LangChain Ecosystem

Module 20: LangChain

Topics

- Chains
- Prompts
- Memory
- Tools
- Agents
- Output Parsers
- Document Loaders
- Retrievers
- RAG Pipelines

Projects

- AI Research Assistant
 - AI SQL Agent
 - AI PDF Chatbot
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Module 21: LangGraph

Topics

- Graph-based AI workflows
- Stateful Agents
- Multi-step AI systems
- Conditional Routing
- Human-in-the-loop
- Agent Memory

Projects

- AI Workflow System
- Autonomous Research Agent
- Multi-Agent AI App

Module 22: LangSmith

Topics

- Observability
- Debugging
- Tracing
- Evaluation
- Monitoring AI workflows

PHASE 11 — AI Agents

Module 23: AI Agents Fundamentals

Topics

- What is AI Agent
- Agent Architecture
- Tools
- Planning
- Memory
- Reflection
- Agent Loops

Agent Types

- Reactive Agents
- Tool-Using Agents
- Autonomous Agents
- Conversational Agents

Module 24: AI Agent Projects

Projects

- AI Travel Agent
- AI Finance Assistant
- AI Coding Assistant
- AI Research Agent
- AI Customer Support Agent

PHASE 12 — Multi-Agent Systems

Module 25: Multi-Agent AI

Topics

- Multi-Agent Systems
- Agent Collaboration
- Agent Communication
- Task Delegation
- Orchestrator Agent
- Supervisor Agent
- Worker Agents

Architectures

- Sequential Agents
- Parallel Agents
- Hierarchical Agents

Projects

- AI Team Simulation
 - AI Startup Assistant
 - AI Recruitment System
 - AI Content Team
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PHASE 13 — Deployment & Cloud

Module 26: Supabase

Topics

- Authentication
 - PostgreSQL
 - Storage
 - Realtime Features
 - Vector Support
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Module 27: Aiven

Topics

- MySQL Hosting
 - PostgreSQL Hosting
 - Cloud Databases
 - Connection Setup
 - Security
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Module 28: Streamlit Cloud

Topics

- Deployment
 - Secrets.toml
 - Environment Variables
 - GitHub Integration
-

Module 29: FastAPI Deployment

Platforms

- Render
- Railway
- VPS
- Docker

Topics

- Uvicorn
 - Gunicorn
 - Docker Basics
 - Environment Variables
-

PHASE 14 — Real-world Projects

Beginner Projects

- Expense Tracker
- Employee CRUD
- Weather App

- Dashboard
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Intermediate Projects

- AI Chatbot
 - AI SQL Assistant
 - PDF Chatbot
 - Resume Analyzer
 - Analytics Dashboard
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Advanced Projects

- Multi-Agent AI System
 - AI SaaS Platform
 - AI Interview Platform
 - Autonomous AI Researcher
 - AI Business Assistant
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Suggested Learning Timeline

Phase	Duration
Python	1 Month
SQL	1 Month
Pandas & Plotly	20 Days
APIs & FastAPI	20 Days
Streamlit	10 Days
AI Fundamentals	15 Days
LLM APIs	15 Days
RAG	15 Days
LangChain Ecosystem	20 Days
AI Agents	20 Days
Deployment	10 Days

Final Skills You Will Have

After completing roadmap:

```
Python Development
SQL Analytics
Data Analysis
Dashboard Building
Backend API Development
LLM Integration
RAG Pipelines
AI Agents
Multi-Agent Systems
Cloud Deployment
AI SaaS Development
```

Final Recommended Stack

```
Frontend:
Streamlit

Backend:
FastAPI

Database:
Supabase / Aiven

AI:
OpenAI / Gemini

RAG:
LangChain + Vector DB

Agents:
LangGraph

Deployment:
Render + Streamlit Cloud
```

Career Roles After This Roadmap

- Python Developer
 - Data Analyst
 - AI Engineer
 - Gen AI Engineer
 - LLM Engineer
 - RAG Engineer
 - AI Agent Developer
 - Backend Developer
 - Full Stack AI Developer
 - AI SaaS Founder
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Final Advice Flow

```
Python
↓
SQL
↓
Pandas + Plotly
↓
FastAPI
↓
Streamlit
↓
LLMs
↓
RAG
↓
LangChain
↓
AI Agents
↓
Multi-Agent Systems
↓
Deployment
```